Uniform Dirichlet boundary bound with a dependent prime-rounding error radius

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Editorial note. The visible joint-model setup names Z_i exactly as the colored mass vector at original1123-1125, with independent iid colors independent of V. It states x>=2 and positive simplex parameters. Theta_x uses that same V; there is no independence assumption between Z_i and Theta_x. This is a setup clarification only; the retained author proof is unchanged.

Editorial note. The event has fixed center u and random radius2Theta_x/log x; Theta_x depends on the same Poisson-Dirichlet sequence that defines Z_i. No independence between Z_i and Theta_x is assumed. The integer coupling N_x,M_x and its old public expected-distance theorem are not needed for this probability-only result.

Editorial note. At1473 the beta integral is printed with reciprocal Gamma product. The preceding Euler identity1454-1455 and the explicit identity1611 uniquely give sin(pi alpha)*Gamma(alpha)*Gamma(1-alpha)=pi. Preserve the original erroneous intermediate display and disclose its reciprocal slip.

Editorial note. In the dyadic display1524, the preceding construction1521 and the goal1529 specify Theta_x-prime>2^m; the displayed unprimed Theta_x and u_i in place of fixed u are notation slips. The rest of the proof consistently controls Theta_x-prime.

Editorial note. The summand at1541 uses V_i while the sum and both surrounding lines1540/1542 run over j. The original sum1510 and dyadic count definition1535 uniquely specify V_j. The author text is retained, not silently altered.

Editorial note. The Campbell formula is used only for nonnegative summands, where its displayed integrability condition applies. This task does not assert the printed unrestricted signed-function version in1817-1823.

Editorial note. The h codomain at469 is written strictly positive although the step sum is0 below lambda_0; the formula472 and explicit observation482 determine the nonnegative rounding function used in r(t)=|h(t)-t|. No numerical value of h is changed.

Editorial scope. One dependent-radius probability bound. The standalone beta integral estimate, gamma/Dirichlet coloring identity and dyadic second-moment lemma are not counted separately. The full integer-factorization Dirichlet theorem is not claimed prepared by this card.

Allowed standard prerequisites (editorial).

- Poisson-Dirichlet gamma-Poisson representation. Descending points of a Poisson process with intensity e^(-t)dt/t, normalized by their sum, have parameter-one Poisson-Dirichlet law.. Represent the joint colored components and the radius on the same point process.
- Poisson coloring, Campbell/Laplace formula and multivariate Mecke equation. Split into independent color processes and remove/reinsert two atoms, preserving their contribution to the random normalization.
- Gamma normalization, Euler reflection and elementary beta integrals. The gamma sum computation and uniform beta-window estimate are printed locally; only standard identities and gamma-to-Dirichlet normalization are free.
- Classical prime number theorem and strong Mertens estimate. Justify the explicit rounding envelope r(t)<=c min(t,t^(-2)) at451-478, not the correlated event estimate.

Source reference. Original Theorem 2: sources/arratia-dirichlet-jep-2025-317/source0.tex, source lines 349–359. The section introduces the probabilistic model of this later theorem. The boundary estimate is proved independently from its colored-PD setup; the integer-factorization theorem is neither an input nor separately counted here.

Joint probability model (editorial setup from the author construction). Fix $x \geq 2$, an integer $k \geq 2$, and positive parameters $\alpha_1, \ldots, \alpha_k$ with $\sum_{i=1}^{k} \alpha_i = 1$. Let $\mathbf{V} = (V_j)_{j\geq 1}$ have the parameter-one Poisson–Dirichlet law. Let $(C_j)_{j\geq 1}$ be independent and identically distributed colors, independent of $\mathbf{V}$, with $\mathbb{P}(C_j = i) = \alpha_i$. Define, on this same probability space,

$$Z_i = \sum_{j:\, C_j = i} V_j \qquad (1 \leq i \leq k).$$

Use the author's Θ_x formed from this very sequence $\mathbf{V}$. Thus Z_i and Θ_x are generally dependent. This setup names the colored masses already defined in the author's Poisson–Dirichlet coloring proposition; it adds no integer-factorization coupling.

There are many ways to define the Poisson–Dirichlet distribution. One of the most intuitive ones involves a "stick-breaking" process that we will use throughout the paper. We start by sampling a sequence of i.i.d. random variables $(U_i)_{i\geq 1}$ that are all uniformly distributed in $[0, 1]$. We then define the sequence $(L_i)_{i\geq 1}$ in the following way:

$$L_1 \coloneqq U_1 \quad \text{and} \quad L_j \coloneqq U_j \prod_{i=1}^{j-1}(1 - U_i) \quad \text{for } j \geq 2.$$

The distribution of the process $\boldsymbol{L} = (L_1, L_2, \ldots)$ is called the *GEM distribution* (of parameter 1), named after Griffiths, Engen and McCloskey (see [10, Ch. 41] for a discussion on the history of this distribution). Lastly, we sort the components of $\boldsymbol{L}$ in non-increasing order to create $\boldsymbol{V} = (V_1, V_2, \ldots)$. The distribution

of this process is the *Poisson–Dirichlet distribution* (of parameter 1).[1] We note that both $\sum_{i\ge1} L_i$ and $\sum_{i\ge1} V_i$ are equal to 1 almost surely.

Let Δ^{k-1} be the set of k-tuples $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k) \in \mathbb{R}^k_{\ge0}$ satisfying $\alpha_1 + \cdots + \alpha_k = 1$. We also need to define a special class of functions:

Starting with the pioneering work of Deshouillers, Dress, and Tenenbaum [7], many different people have studied the distribution of $\boldsymbol{D}_{f,x}$ for various choices of $f \in \mathcal{F}_k(\boldsymbol{\alpha})$, and they proved that it converges on a logarithmic scale to a certain *Dirichlet distribution*. Recall that if $\boldsymbol{\alpha} \in \Delta^{k-1}$ satisfies $\alpha_i > 0$ for all i, we say that a Δ^{k-1}-valued random vector $\boldsymbol{Z}$ follows the Dirichlet distribution $\mathrm{Dir}(\boldsymbol{\alpha})$ if

$$\mathbb{P}\big[Z_i \le u_i \ \ \forall i < k\big] = F_{\boldsymbol{\alpha}}(\boldsymbol{u}) \coloneqq \prod_{i=1}^{k} \Gamma(\alpha_i)^{-1} \int\limits_{\substack{0\le t_i\le u_i \ \forall i<k \\ t_1+\cdots+t_{k-1}\le 1}} \cdots \int \prod_{i=1}^{k} t_i^{\alpha_i-1}\, \mathrm{d}t_1 \cdots \mathrm{d}t_{k-1}.$$

Let $(\lambda_j)_{j\ge0}$ be the increasing sequence of positive real numbers defined by $\lambda_0 \coloneqq e^{-\gamma}$ and

$$\lambda_j \coloneqq \exp\Big(-\gamma + \sum_{i\le j} \frac{1}{v_i q_i}\Big) \quad \text{for } j \ge 1,$$

with γ being the Euler-Mascheroni constant and $q_j = p_j^{v_j}$ being the jth smallest prime power, i.e. $(q_j)_{j\ge1}$ is the sequence $2, 3, 2^2, 5, 7, 2^3, 3^2, \dots$ Note that

$$\lambda_j = \log q_j + O(1/(\log q_j)^2) \tag{1}$$

by a stronger version of Mertens's estimate (Proposition 0.2).

Moreover, we have

$$q_{j+1} = q_j + O(q_j/(\log q_j)^3). \tag{2}$$

Indeed, the prime number theorem (Proposition 0.1) implies that for some sufficiently large constant C, we have $\theta(q_j + Cq_j/(\log q_j)^3) - \theta(q_j) > 0$, whence $q_{j+1} \le q_j + Cq_j/(\log q_j)^3$, as required.

Next, we define the step-function $h : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ by

$$h(t) \coloneqq \sum_{j\ge1} (\log q_j) \cdot \mathbb{1}_{\lambda_{j-1} < t \le \lambda_j}. \tag{3}$$

In particular, (1) and (2) imply that, if $r(t) \coloneqq |h(t) - t|$, then

$$r(t) \ll \min\{t, t^{-2}\} \quad \text{for all } t > 0. \tag{4}$$

$$\Theta_x \coloneqq \sum_{i\ge1} r(V_i \log x). \tag{5}$$

$$\mathcal{E}_{\mathrm{PD}}(\Delta^{k-1}_{\mathbf{u}})^c \subseteq \bigcup_{\substack{i<k \\ u_i\ne1}} \mathcal{B}^{(i)}_{\mathrm{PD}}(u_i), \quad \text{where} \quad \mathcal{B}^{(i)}_{\mathrm{PD}}(u) \coloneqq \Big\{\big|Z_i - u\big| \le {}^{2\Theta_x}\!/_{\log x}\Big\},$$

[1] The GEM and Poisson–Dirichlet distributions have more general definitions involving typically a parameter θ. In the rest of the paper, we will not be mentioning the parameter since we will always work with $\theta = 1$.

Proposition 0.1 (The prime number theorem)**.** *For $x \geq 2$, we have*

$$\theta(x) = x + O(x/(\log x)^3)$$

and

$$\psi(x) = x + O(x/(\log x)^3).$$

We shall also need a stronger version of Mertens's estimate, which in its classical form, yields an estimate for the product $\prod_{p \leq x}(1 - 1/p)$.

Proposition 0.2 (Strong Mertens's estimate)**.** *For $x \geq 2$, we have*

$$\sum_{p^k \leq x} \frac{1}{kp^k} = \log_2 x + \gamma + O(1/(\log x)^3).$$

Proof. Using the prime number theorem (Proposition 0.1), there exists a constant c_0 satisfying

$$\sum_{p^k \leq x} \frac{1}{kp^k} = \int_{2^-}^{x} \frac{\mathrm{d}\psi(t)}{t \log t} = \log_2 x + c_0 + O(1/(\log x)^3).$$

for $x \geq 2$. For the proof of $\gamma = c_0$, see the proof of [17, Th. 2.7]. □

0.1 Poisson Processes

The following definition and propositions are borrowed from Kingman's book on Poisson processes [11]. Let $(S, \mathcal{S})$ be a measurable space with S being a subset of $\mathbb{R}^d$ for some $d \geq 1$. A *Poisson process* on a state space S with mean measure μ is a random countable subset $\Pi \subseteq S$ such that:

- for any disjoint measurable subsets $A_1, \ldots, A_n$ of S, the random variables $\#(\Pi \cap A_1)$, ..., $\#(\Pi \cap A_n)$ are independent;
- the random variable $\#(\Pi \cap A)$ is a Poisson random variable of parameter $\mu(A)$ for any $A \subset S$ measurable.

We can see Π as an element of the measurable space $(\Omega_S, \mathcal{F}_S)$ where Ω_S is the set of countable subsets of S and $\mathcal{F}_S$ is the smallest σ-algebra for which the map $\Pi \mapsto \#\{\Pi \cap B\}$ is measurable for all $B \in \mathcal{S}$. If μ has no atoms, meaning no singleton with positive probability, and is σ-finite, meaning that S is a countable union of measurable sets with finite measure, then a Poisson process with mean measure μ always exists (see [11], the existence theorem in section 2.5 for a proof). If μ is absolutely continuous with respect to the Lebesgue measure, then the function $\lambda\colon S \to \mathbb{R}_{\geq 0}$ such that $\mu(A) = \int_A \lambda(x)\,\mathrm{d}x$ for all measurable subsets $A \subseteq S$ is called the *intensity* of the Poisson process. Here are a few important propositions about Poisson processes that we will use in the paper and proved in [11]:

Proposition 0.3 (Mapping theorem, [11, §2.3])**.** *If Π is a Poisson process with mean measure μ on S, and $f : S \to T$ is a measurable function such that $\mu^*(B) \coloneqq \mu(f^{-1}(B))$ has no atoms, then $f(\Pi)$ is a Poisson process on T with measure μ^*.*

Proposition 0.4 (Coloring theorem, [11, §5.1]). *If Π is a Poisson process with mean measure μ on S, and the points are randomly colored with k colors such that the probability of a point receiving the color i is p_i, and such that the color of a point is independent of different points and of the position of the point. Let Π_i be the subset of Π with color i. Then all the Π_i are independent Poisson processes with mean measures $\mu_i = p_i\mu$.*

Proposition 0.5 (Campbell's theorem, [11, §3.2]). *Let Π be any Poisson process on S with mean measure μ. Let $f\colon S \to \mathbb{R}$ be a measurable function. Then*

$$\Sigma = \sum_{X\in\Pi} f(X)$$

is absolutely convergent almost surely if, and only if,

$$\int_S \min\{f, 1\}\,\mathrm{d}\mu < \infty.$$

If this condition holds, then

$$\mathbb{E}[e^{s\Sigma}] = \exp\left(\int_S (e^{sf} - 1)\,\mathrm{d}\mu\right)$$

for any complex s for which the integral converges. Moreover,

$$\mathbb{E}[\Sigma] = \int_S f\,\mathrm{d}\mu$$

if the integral converges. In the case where it converges, we also have

$$\mathrm{Var}[\Sigma] = \int_S f^2\,\mathrm{d}\mu.$$

Many probabilities or expectations that involve Poisson processes in this paper can be reformulated as

$$\mathbb{E}\sum_{X\in\Pi} f(\Pi\setminus\{X\}, X).$$

In these cases, there is a generalization of the formula for $\mathbb{E}[\Sigma]$ in Campbell's theorem, called the Mecke equation, allowing us to compute these objects:

Proposition 0.6 (Mecke equation, [14, Th. 4.5]). *Let Π be a Poisson process on S with a σ-finite mean measure μ, and let $f : \Omega_S \times S^k \to [0,\infty)$ be measurable. Then we have*

$$\mathbb{E}\sum_{\substack{X_1,\dots,X_k\\ \text{all distinct}}} f(\Pi\setminus\{X_1,\dots,X_k\}; X_1,\dots,X_k) = \int_S\cdots\int_S \mathbb{E}[f(\Pi; x_1,\dots,x_k)]\,\mathrm{d}\mu(x_1)\cdots\mathrm{d}\mu(x_k).$$

Lemma 0.7. *For $i \in \{1, 2, \dots, k-1\}$ and $u \in (0,1)$, we have*

$$\mathbb{P}\big[\mathcal{B}^{(i)}_{\mathrm{PD}}(u)\big] \ll \frac{1}{(1+u\log x)^{1-\alpha_i}(1+(1-u)\log x)^{\alpha_i}}.$$

1 Theorem 2 for the probabilistic model

With the following probabilistic version of Theorem 2 given by Donnelly and Tavaré in 1987 [8, §3], we see how a Poisson–Dirichlet process relates to the Dirichlet distribution. We repeat their proof since they use a Poisson process that will come in handy later.

Proposition 1.1 (Donnelly–Tavaré [8])**.** *Let $\boldsymbol{V} = (V_1, V_2, \ldots)$ be a Poisson–Dirichlet process, let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_k) \in \Delta^{k-1}$ such that $\alpha_i > 0$ for all $1 \leq i \leq k$, and let $(C_i)_{i \geq 1}$ be a sequence of i.i.d. random variables (also independent of $\boldsymbol{V}$) with $\mathbb{P}[C_i = j] = \alpha_j$ for all $i \geq 1$. Then the random vector*

$$\Big(\sum_{i:\ C_i = 1} V_i, \ldots, \sum_{i:\ C_i = k} V_i \Big)$$

is distributed according to $\mathrm{Dir}(\boldsymbol{\alpha})$.

Proof. Let $X_1 > X_2 > \cdots$ be the points of a Poisson process on $\mathbb{R}_{>0}$ with intensity $(e^{-x}/x)\,\mathrm{d}x$, and let $S \coloneqq \sum_{i \geq 1} X_i$. For each $i \geq 1$, Let $V_i \coloneqq X_i/S$. We then know that $(V_1, V_2, \ldots)$ follows the Poisson–Dirichlet distribution [9, Th. 2.2]. With the coloring theorem (Proposition 0.4), the point processes

$$\Pi_m = \{X_i : C_i = m\}$$

form independent Poisson processes of intensity $\alpha_m \cdot (e^{-x}/x)\,\mathrm{d}x$ for all $m = 1, \ldots, k$. We then let

$$S_m \coloneqq \sum_{X \in \Pi_m} X$$

for $m = 1, 2, \ldots, k$. With Campbell's theorem (Proposition 0.5), we compute the moment generating function

$$\log \mathbb{E}[e^{sS_m}] = \alpha_m \cdot \int_0^\infty \frac{e^{sx} - 1}{x} \cdot e^{-x}\,\mathrm{d}x = -\alpha_m \log(1 - s)$$

for $\mathrm{Re}(s) < 1$. This coincides with the moment generating function of $\mathrm{Gamma}(\alpha_m, 1)$ distribution. We thus deduce that the vector

$$\Big(\sum_{i:\ C_i = 1} V_i, \ldots, \sum_{i:\ C_i = k} V_i \Big) = \Big(\frac{S_1}{\sum_{m=1}^k S_m}, \ldots, \frac{S_k}{\sum_{m=1}^k S_m} \Big)$$

follows the distribution $\mathrm{Dir}(\boldsymbol{\alpha})$ (see [12, Ch. 49, pp. 485-487] for a proof). □

2 Proof of Lemma 0.7 using probabilistic techniques

Since $\boldsymbol{Z}$ follows the distribution $\mathrm{Dir}(\boldsymbol{\alpha})$, each component Z_i follows a $\mathrm{Beta}(\alpha_i, 1-\alpha_i)$ distribution, meaning that if $[a, b] \subseteq [0, 1]$, then

$$\begin{aligned} \mathbb{P}\big[Z_i \in [a, b]\big] &= \frac{1}{\Gamma(\alpha_i)\Gamma(1 - \alpha_i)} \int_a^b \frac{\mathrm{d}t}{t^{1-\alpha_i}(1-t)^{\alpha_i}} \\ &= \frac{\sin(\pi\alpha_i)}{\pi} \int_a^b \frac{\mathrm{d}t}{t^{1-\alpha_i}(1-t)^{\alpha_i}}, \end{aligned} \tag{6}$$

by Euler's reflection formula. For this reason, we need the following preliminary estimate.

Lemma 2.1. *Uniformly for $u \in [0,1]$, $\alpha \in (0,1)$ and $\delta > 0$, we have*

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}(1-u+\delta)^{\alpha}}.$$

In particular, if $\delta \geq 1/\log x$, then

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} \ll \frac{\delta \log x}{(1+u\log x)^{1-\alpha}(1+(1-u)\log x)^{\alpha}}.$$

Proof. Note that both sides of the claimed inequality are invariant under the change of variables $(\alpha, u) \mapsto (1-\alpha, 1-u)$. Hence, we may assume without loss of generality that $u \in [0, 1/2]$. In addition, observe that

$$\sin(\pi\alpha) \int_0^1 \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}} = \frac{\sin(\pi\alpha)}{\Gamma(\alpha)\Gamma(1-\alpha)} = \pi.$$

Hence the lemma is trivially true if $\delta > 1/4$.

We have thus reduced the proof to the case when $\alpha \in (0,1)$, $u \in [0,1/2]$ and $\delta \in [0,1/4]$. In particular, $u+\delta \leq 3/4$, so that $1-t \in [1/4,1]$ for all $t \in [0, u+\delta]$. It thus suffices to prove that

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}}. \tag{7}$$

Assume first that $\delta \leq u/2$. We then have $t \geq u/2$ whenever $t \in [u-\delta, u+\delta]$. Using also the trivial bound $\sin(\pi\alpha) \leq 1$, we conclude that

$$\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} \leq \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{(u/2)^{1-\alpha}} \ll \frac{\delta}{(u+\delta)^{1-\alpha}}.$$

This proves the lemma in this case.

Finally, assume that $\delta \geq u/2$. We then use that

$$\begin{aligned}\sin(\pi\alpha) \int_{[u-\delta,u+\delta]\cap[0,1]} \frac{\mathrm{d}t}{t^{1-\alpha}} &\leq \sin(\pi\alpha) \int_0^{3\delta} \frac{\mathrm{d}t}{t^{1-\alpha}} = \frac{\sin(\pi\alpha)}{\alpha} \cdot (3\delta)^{\alpha} \\ &\ll \delta^{\alpha} \asymp \frac{\delta}{(u+\delta)^{1-\alpha}}.\end{aligned}$$

This completes the proof of the lemma in all cases. □

Let us now show Lemma 0.7. For simplicity of notation, let us fix $i \in \{1,2,\dots,k\}$ and $u \in (0,1)$, and let us set $\alpha := \alpha_i$ and

$$\Delta := \frac{1}{(1+u\log x)^{1-\alpha}(1+(1-u)\log x)^{\alpha}}.$$

Thus, our goal is to show that

$$\mathbb{P}\Big[\big|Z_i - u\big| \leq {}^{2\Theta_x}/_{\log x}\Big] \ll \Delta. \tag{8}$$

Recall that

$$\Theta_x = \sum_{j\ge 1} r(V_j \log x),$$

and that there exists an absolute constant $c \ge 1$ such that $r(y) \le c\min\{y, y^{-2}\}$ for all $y > 0$ (see (4)). Since there are at most 10 indices j such that $V_j \ge 0.1$, we find

$$\Theta_x \le \Theta'_x + 10c, \quad \text{where} \quad \Theta'_x \coloneqq \sum_{j\ge 1:\ V_j<0.1} r(V_j \log x).$$

Now, using Lemma 2.1 and relation (6), we have

$$\mathbb{P}\Big[\big|Z_i - u\big| \le {}^{200c}/_{\log x}\Big] \ll \Delta.$$

On the other hand, if $\frac{200c}{\log x} < |Z_i - u| \le \frac{2\Theta_x}{\log x} \le \frac{20c+2\Theta'_x}{\log x}$, then we must have $\Theta'_x > 90c$. In particular, there exists $m \in \mathbb{Z}$ such that $2^m > 40c$ and $\Theta'_x \in (2^m, 2^{m+1}]$, whence $|Z_i - u| \le \frac{20c+2^{m+2}}{\log x}$. We thus conclude that

$$\mathbb{P}\Big[\big|Z_i - u\big| \le \frac{20c + 2\Theta'_x}{\log x}\Big] \le \sum_{m\in\mathbb{Z}:\ 2^m>40c} \mathbb{P}\Big[\big|Z_i - u_i\big| \le \frac{20c + 2^{m+2}}{\log x},\ \Theta_x > 2^m\Big] + O(\Delta).$$

Therefore Lemma 0.7 will follow if we can prove that

$$\mathbb{P}\Big[\big|Z_i - u\big| \le \frac{20c + 4\kappa}{\log x},\ \Theta'_x > \kappa\Big] \ll \frac{\Delta}{\kappa} \qquad \text{for all } \kappa \ge 40c. \tag{9}$$

We shall make a further reduction. If we set

$$G(\lambda) \coloneqq \#\big\{j \ge 1 : V_j \log x \in (\lambda, 2\lambda]\big\},$$

then we have

$$\begin{aligned}
\Theta'_x = \sum_{j\ge 1:\ V_j<0.1} r(V_j \log x) &\le c \sum_{j\ge 1:\ V_j<0.1} \min\big\{V_i \log x, (V_i \log x)^{-2}\big\} \\
&\le c \sum_{\substack{m\ge 0 \\ 2^m \le 0.1 \log x}} \frac{G(2^m)}{4^m} + c\sum_{m<0} 2^{m+1} G(2^m) \\
&\le 5c \max_{\substack{m\ge 0 \\ 2^m \le 0.1 \log x}} \Big(\frac{G(2^m)}{2^{3m/2}}\Big) + 5c \max_{m<0} \big(2^{m/2} G(2^m)\big),
\end{aligned}$$

since $\sum_{m\ge 0} 2^{-m/2} < 5$ and $\sum_{m<0} 2^{1+m/2} < 5$. We thus find

$$\begin{aligned}
&\mathbb{P}\Big[\big|Z_i - u\big| \le \frac{20c + 4\kappa}{\log x},\ \Theta'_x > \kappa\Big] \\
&\qquad \le \sum_{\substack{m\ge 0 \\ 2^m \le 0.1 \log x}} \mathbb{P}\Big[\big|Z_i - u\big| \le \frac{20c + 4\kappa}{\log x},\ G(2^m) > \frac{2^{3m/2}\kappa}{10c}\Big] \\
&\qquad\qquad + \sum_{m<0} \mathbb{P}\Big[\big|Z_i - u\big| \le \frac{20c + 4\kappa}{\log x},\ G(2^m) > \frac{\kappa 2^{-m/2}}{10c}\Big].
\end{aligned}$$

Note that in both sums, we have $G(2^m) \geq 4$, since $\kappa \geq 40c$. Hence, Markov's inequality implies

$$\begin{aligned}\mathbb{P}\Big[\big|Z_i - u\big| \leq \frac{20c+4\kappa}{\log x},\ \Theta'_x > \kappa\Big] \\ \leq \sum_{\substack{m\geq 0\\ 2^m \leq 0.1\log x}} \frac{100c^2}{\kappa^2 2^{3m}}\mathbb{E}\Big[\mathbb{1}_{|Z_i-u|\leq \frac{20c+4\kappa}{\log x}} \cdot \mathbb{1}_{G(2^m)\geq 4}\cdot G(2^m)^2\Big] \\ + \sum_{m<0} \frac{100c^2 2^m}{\kappa^2}\mathbb{E}\Big[\mathbb{1}_{|Z_i-u|\leq \frac{20c+4\kappa}{\log x}} \cdot \mathbb{1}_{G(2^m)\geq 4}\cdot G(2^m)^2\Big].\end{aligned}$$

This reduces (9), and thus Lemma 0.7, to proving the following estimate:

Lemma 2.2. *Uniformly for $\mu \geq 1$ and $\lambda \in (0, 0.1\log x]$, we have*

$$\mathbb{E}\Big[\mathbb{1}_{|Z_i-u|\leq \mu/\log x} \cdot \mathbb{1}_{G(\lambda)\geq 4}\cdot G(\lambda)^2\Big] \ll (\lambda+\mu)\cdot\Delta.$$

Proof. Let us call E the quantity we seek to bound from above. Consider two independent Poisson processes $(A_j)_{j\geq 1}$ with intensity $\alpha(e^{-x}/x)\,\mathrm{d}x$ and $(B_j)_{j\geq 1}$ with intensity $(1-\alpha)(e^{-x}/x)\,\mathrm{d}x$. Hence, the union of the two processes is a new Poisson process of intensity $(e^{-x}/x)\,\mathrm{d}x$. The sum $S_A \coloneqq \sum_{j\geq 1} A_j$ has distribution Gamma$(\alpha, 1)$ and the sum $S_B \coloneqq \sum_{j\geq 1} B_j$ has distribution Gamma$(1-\alpha, 1)$. If we also set $S \coloneqq S_A + S_B$, then, as we saw in the proof of Proposition 1.1, we have

$$E = \mathbb{E}\Big[\mathbb{1}_{|S_A/S-u|\leq \mu/\log x}\cdot \mathbb{1}_{G_A(\lambda)+G_B(\lambda)\geq 4}\cdot \big(G_A(\lambda)+G_B(\lambda)\big)^2\Big],$$

where $G_A(\lambda) = \sum_{j\geq 1}\mathbb{1}_{A_j\log x\in(\lambda S, 2\lambda S]}$ and $G_B(\lambda) = \sum_{j\geq 1}\mathbb{1}_{B_j\log x\in(\lambda S,2\lambda S]}$. We know $G_A(\lambda)+G_B(\lambda)\geq 4$. Hence, if $G_A(\lambda)\geq G_B(\lambda)$, then we must have $G_A(\lambda)\geq 2$ and $G_A(\lambda)\leq G_A(\lambda)^2/2$, whence $G_A(\lambda)\leq 2\sum\sum_{k>j\geq 1}\mathbb{1}_{A_k\log x,\, A_j\log x\in(\lambda S,2\lambda S]}$. So, we find

$$\begin{aligned}\big(G_A(\lambda)+G_B(\lambda)\big)^2 \leq 4G_A(\lambda)^2 = 4G_A(\lambda) + 8\sum_{k>j\geq 1}\sum \mathbb{1}_{A_k\log x,\, A_j\log x\in(\lambda S,2\lambda S]} \\ \leq 16\sum_{k>j\geq 1}\sum \mathbb{1}_{A_k\log x,\, A_j\log x\in(\lambda S,2\lambda S]}.\end{aligned}$$

The analogous inequality also holds when $G_B(\lambda)\geq G_A(\lambda)$, with the roles of A and B reversed. We conclude that

$$E \leq 16(E_A + E_B),$$

where

$$E_A \coloneqq \mathbb{E}\Big[\mathbb{1}_{|S_A/S-u|\leq\mu/\log x}\sum\sum_{k>j\geq 1}\mathbb{1}_{A_k\log x,\, A_j\log x\in(\lambda S,2\lambda S]}\Big]$$

and E_B is defined analogously. Using Mecke's equation (cf. Proposition 0.6),

we find

$$E_A = \alpha^2 \iint\limits_{2a_1>a_2>a_1>0} \mathbb{E}\Big[\mathbb{1}_{|\frac{a_1+a_2+S_A}{a_1+a_2+S}-u|\le\frac{\mu}{\log x}} \prod_{j=1}^{2} \mathbb{1}_{a_j\log x\in(\lambda(a_1+a_2+S),2\lambda(a_1+a_2+S))}\Big] \times \frac{e^{-a_1-a_2}}{a_1a_2}\,\mathrm{d}a_1\,\mathrm{d}a_2$$

$$= \frac{\alpha^2}{\Gamma(\alpha)\Gamma(1-\alpha)} \iiiint\limits_{\substack{2a_1>a_2>a_1>0,\ s_1,s_2>0\\ |\frac{a_1+a_2+s_1}{a_1+a_2+s_1+s_2}-u|\le\frac{\mu}{\log x}\\ a_j\log x\in(\lambda(a_1+a_2+s_1+s_2),2\lambda(a_1+a_2+s_1+s_2))\ (j=1,2)}} \frac{e^{-a_1-a_2-s_1-s_2}}{a_1a_2s_1^{1-\alpha}s_2^{\alpha}}\,\mathrm{d}a_1\,\mathrm{d}a_2\,\mathrm{d}s_1\,\mathrm{d}s_2,$$

where we used that the a_j's lie in the same dyadic interval to deduce that $a_2 < 2a_1$. We make the change of variables $t = s_1/(s_1+s_2)$ and $s = s_1+s_2$. Since $\lambda/\log x \le 0.1$ and $a_1 < a_2 < 2a_1$, the conditions $a_j\log x \in (\lambda(a_1+a_2+s), 2\lambda(a_1+a_2+s))$ for $j=1,2$ imply that $a_j\log x \in (\lambda s, 5\lambda s]$. Knowing also that $|\frac{a_1+a_2+s_1}{a_1+a_2+s}-u| \le \frac{\mu}{\log x}$, we find $|t-u| = |\frac{s_1}{s_1+s_2}-u| \le \frac{10\lambda+\mu}{\log x}$. Finally, we use Euler's reflection formula to write $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin(\pi\alpha)$. We conclude that

$$E_A \le \frac{\sin(\pi\alpha)}{\pi} \iiiint\limits_{\substack{a_2>a_1>0,\ s>0,\ t\in[0,1]\\ |t-u|\le(10\lambda+\mu)/\log x\\ a_j\log x\in(\lambda s,5\lambda s]\ (j=1,2)}} \frac{e^{-a_1-a_2-s}}{a_1a_2t^{1-\alpha}(1-t)^{\alpha}}\,\mathrm{d}a_1\,\mathrm{d}a_2\,\mathrm{d}s\,\mathrm{d}t.$$

For every fixed value of s, the integral over a_1 and a_2 is $\le (\log 5)^2$, since $\int_w^{5w}\frac{e^{-a}}{a}\,\mathrm{d}a \le \log 5$ for any $w>0$. We also have $\int_0^{\infty} e^{-s}\,\mathrm{d}s = 1$. We thus conclude that

$$E_A \le \frac{(\log 5)^2\sin(\pi\alpha)}{\pi} \int\limits_{\substack{t\in[0,1]\\ |t-u|\le(10\lambda+\mu)/\log x}} \frac{\mathrm{d}t}{t^{1-\alpha}(1-t)^{\alpha}}.$$

Using Lemma 2.1 shows that $E_A \ll (\lambda+\mu)\Delta$. The same estimate holds for E_B too, thus completing the proof of the lemma. □

References

[1] R. Arratia - "On the central role of scale invariant Poisson processes on (0,∞)", in Microsurveys in discrete probability (Princeton, NJ, 1997), DIMACS Ser. Discrete Math. Theoret. Comput. Sci., vol. 41, American Mathematical Society, Providence, RI, 1998, p. 21-41

[2] R. Arratia - "On the amount of dependence in the prime factorization of a uniform random integer", in Contemporary combinatorics, Bolyai Soc. Math. Stud., vol. 10, János Bolyai Math. Soc., Budapest, 2002, p. 29-91

[3] R. Arratia, A. D. Barbour & S. Tavaré - "A tale of three couplings: Poisson-Dirichlet and GEM approximations for random permutations", Combin. Probab. Comput. 15 (2006) no. 1-2, p. 31-62

[4] G. Bareikis & E. Manstavičius - "On the DDT theorem", Acta Arith. 126 (2007) no. 2, p. 155-168

[5] P. Billingsley - "On the distribution of large prime divisors", Period. Math. Hungar. 2 (1972), p. 283-289

[6] R. de la Bretèche & G. Tenenbaum - "Sur les processus arithmétiques liés aux diviseurs", Adv. in Appl. Probab. 48 (2016) no. A, p. 63-76

[7] J.-M. Deshouillers, F. Dress & G. Tenenbaum - "Lois de répartition des diviseurs. I", Acta Arith. 34 (1979) no. 4, p. 273-285 (loose errata)

[8] P. Donnelly & S. Tavaré - "The population genealogy of the infinitely-many neutral alleles model", J. Math. Biol. 25 (1987) no. 4, p. 381-391

[9] S. Feng - The Poisson-Dirichlet distribution and related topics. Models and asymptotic behaviors, Probability and its Appl., Springer, Heidelberg, 2010

[10] N. L. Johnson, S. Kotz & N. Balakrishnan - Discrete multivariate distributions, Wiley Series in Probab. and Stat.: Applied Probab. and Stat., John Wiley & Sons, Inc., New York, 1997

[11] J. F. C. Kingman - Poisson processes, Oxford Studies in Probab., vol. 3, The Clarendon Press, Oxford University Press, New York, 1993

[12] S. Kotz, N. Balakrishnan & N. L. Johnson - Continuous multivariate distributions. Vol. 1. Models and applications, Wiley Series in Probab. and Stat.: Applied Probab. and Stat., Wiley-Interscience, New York, 2000

[13] D. Koukoulopoulos - The distribution of prime numbers, Graduate Studies in Math., vol. 203, American Mathematical Society, Providence, RI, 2019

[14] G. Last & M. Penrose - Lectures on the Poisson process, Institute of Math. Stat. Textbooks, vol. 7, Cambridge University Press, Cambridge, 2018

[15] S.-K. Leung - "Dirichlet law for factorisation of integers, polynomials and permutations", Math. Proc. Cambridge Philos. Soc. 175 (2023) no. 3, p. 649-676

[16] S.-K. Leung - "Dirichlet law with arbitrary parameters for factorizations of integers" (2025), Retrieved on 24 August 2025 from https://sites.google.com/view/sunkaileung/miscellaneous

[17] H. L. Montgomery & R. C. Vaughan - Multiplicative number theory. I. Classical theory, Cambridge Studies in Advanced Math., vol. 97, Cambridge University Press, Cambridge, 2007

[18] S. Nyandwi & A. Smati - "Distribution laws of pairs of divisors", Integers 13 (2013), article ID A13, 13 pages

[19] G. Tenenbaum - "A rate estimate in Billingsley's theorem for the size distribution of large prime factors", Q. J. Math. 51 (2000) no. 3, p. 385-403